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1. Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

```
HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
  Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
  Server: Apache/2.0.40 (Red Hat Linux)\r\n
  Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n
  ETag: "1bfed-49-79d5bf00"\r\n
  Accept-Ranges: bytes\r\n
  Content-Length: 73\r\n
  Keep-Alive: timeout=10, max=100\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=ISO-8859-1\r\n
  \r\n
  [Request in frame: 10]
  [Time since request: 24.143000 milliseconds]
  [Request URI: /ethereal-labs/lab2-1.html]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
  File Data: 73 bytes
Line-based text data: text/html (3 lines)
```

```
Hypertext Transfer Protocol
  GET /ethereal-labs/lab2-1.html HTTP/1.1\r\n
      Request Method: GET
      Request URI: /ethereal-labs/lab2-1.html
      Request Version: HTTP/1.1
  Host: gaia.cs.umass.edu\r\n
  User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
  Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,text/css,*/*;…
  Accept-Language: en-us, en;q=0.50\r\n
  Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
  Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
  Keep-Alive: 300\r\n
  Connection: keep-alive\r\n
  \r\n
  [Response in frame: 12]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
```

Both my browser and the server is running on HTTP version 1.1

2. What languages (if any) does your browser indicate that it can accept to the server?

Accepted language is English. "en-us, en;q=0.50"

```
Hypertext Transfer Protocol
  GET /ethereal-labs/lab2-1.html HTTP/1.1\r\n
      Request Method: GET
      Request URI: /ethereal-labs/lab2-1.html
      Request Version: HTTP/1.1
  Host: gaia.cs.umass.edu\r\n
  User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
  Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,text/css,*/*;…
  Accept-Language: en-us, en;q=0.50\r\n
  Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
  Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
  Keep-Alive: 300\r\n
  Connection: keep-alive\r\n
  \r\n
  [Response in frame: 12]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
```

3. What is the IP address of your computer? Of the gaia.cs.umass.edu server?

10	4.694850	192.168.1.102	128.119.245.12	HTTP	555 GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439 HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541 GET /favicon.ico HTTP/1.1
14	4.758366	128.119.245.12	192.168.1.102	HTTP	1395 HTTP/1.1 404 Not Found (text/html)

Source is mine: 192.168.1.102

Destination is server: 128.119.245.12

4. What is the status code returned from the server to your browser?

```
10 4.694850   192.168.1.102   128.119.245.12   HTTP   555 GET /ethereal-labs/lab2-1.html HTTP/1.1
12 4.718993   128.119.245.12  192.168.1.102    HTTP   439 HTTP/1.1 200 OK  (text/html)
13 4.724332   192.168.1.102   128.119.245.12   HTTP   541 GET /favicon.ico HTTP/1.1
14 4.750366   128.119.245.12  192.168.1.102    HTTP  1395 HTTP/1.1 404 Not Found  (text/html)
```

HTTP 200 OK.

5. When was the HTML file that you are retrieving last modified at the server?

```
▾ HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
  Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
  Server: Apache/2.0.40 (Red Hat Linux)\r\n
  Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n
  ETag: "1bfed-49-79d5bf00"\r\n
  Accept-Ranges: bytes\r\n
▸ Content-Length: 73\r\n
  Keep-Alive: timeout=10, max=100\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=ISO-8859-1\r\n
  \r\n
  [Request in frame: 10]
  [Time since request: 24.143000 milliseconds]
  [Request URI: /ethereal-labs/lab2-1.html]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
  File Data: 73 bytes
▸ Line-based text data: text/html (3 lines)
```

Date: Tue, 23 Sep 2003 05.29

6. How many bytes of content are being returned to your browser?

```
▾ HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
  Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
  Server: Apache/2.0.40 (Red Hat Linux)\r\n
  Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n
  ETag: "1bfed-49-79d5bf00"\r\n
  Accept-Ranges: bytes\r\n
▸ Content-Length: 73\r\n
  Keep-Alive: timeout=10, max=100\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=ISO-8859-1\r\n
  \r\n
  [Request in frame: 10]
  [Time since request: 24.143000 milliseconds]
  [Request URI: /ethereal-labs/lab2-1.html]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
  File Data: 73 bytes
▸ Line-based text data: text/html (3 lines)
```

File Data: 73 bytes

7. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

```
▾ GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /ethereal-labs/lab2-2.html
    Request Version: HTTP/1.1
  Host: gaia.cs.umass.edu\r\n
  User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
  Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,text/css,*/*;q=
  Accept-Language: en-us, en;q=0.50\r\n
  Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
  Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
  Keep-Alive: 300\r\n
  Connection: keep-alive\r\n
  \r\n
  [Response in frame: 10]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
```

No, there is no content about modified or not.

8. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

```
▾ HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
  Date: Tue, 23 Sep 2003 05:35:50 GMT\r\n
  Server: Apache/2.0.40 (Red Hat Linux)\r\n
  Last-Modified: Tue, 23 Sep 2003 05:35:00 GMT\r\n
  ETag: "1bfef-173-8f4ae900"\r\n
  Accept-Ranges: bytes\r\n
▾ Content-Length: 371\r\n
    [Content length: 371]
  Keep-Alive: timeout=10, max=100\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=ISO-8859-1\r\n
  \r\n
  [Request in frame: 8]
  [Time since request: 26.634000 milliseconds]
  [Request URI: /ethereal-labs/lab2-2.html]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
  File Data: 371 bytes
▸ Line-based text data: text/html (10 lines)
```

Yes. It says 200 OK. Content – Type: text/html. This section displays the actual HTML source code of the requested page.

9. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

```
▾ GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /ethereal-labs/lab2-2.html
    Request Version: HTTP/1.1
  Host: gaia.cs.umass.edu\r\n
  User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
  Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,text/css,*/*
  Accept-Language: en-us, en;q=0.50\r\n
  Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
  Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
  Keep-Alive: 300\r\n
  Connection: keep-alive\r\n
  If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n
  If-None-Match: "1bfef-173-8f4ae900"\r\n
  Cache-Control: max-age=0\r\n
  \r\n
  [Response in frame: 15]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
```

Yes. There is a section says If-Modified-Since: "Tue, 23 Sep 2003 05:35:00 GMT" the date modified.

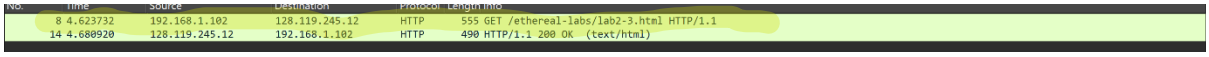
10. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

```
▾ Hypertext Transfer Protocol
  ▾ HTTP/1.1 304 Not Modified\r\n
      Response Version: HTTP/1.1
      Status Code: 304
      [Status Code Description: Not Modified]
      Response Phrase: Not Modified
    Date: Tue, 23 Sep 2003 05:35:53 GMT\r\n
    Server: Apache/2.0.40 (Red Hat Linux)\r\n
    Connection: Keep-Alive\r\n
    Keep-Alive: timeout=10, max=99\r\n
    ETag: "1bfef-173-8f4ae900"\r\n
    \r\n
    [Request in frame: 14]
    [Time since request: 22.826000 milliseconds]
    [Request URI: /ethereal-labs/lab2-2.html]
    [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
```

304 Not Modified and no. The HTTP status code is '304 Not Modified'. T server did not explicitly return the contents of the file. Because my second GET request included the 'If-

Modified-Since' header and the file had not changed on the server, the server simply responded with a 304 code. This instructs the browser to load the page from its local cache, saving bandwidth.

11. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill or Rights?



No.	Time	Source	Destination	Protocol	Length	Info
8	4.623732	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-3.html HTTP/1.1
14	4.680920	128.119.245.12	192.168.1.102	HTTP	490	HTTP/1.1 200 OK (text/html)

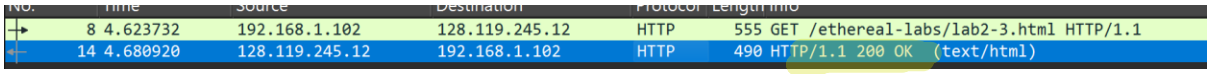
One get requests. The packet no 8.

12. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

No.	Time	Source	Destination	Protocol	Length	Info
8	4.623732	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-3.html HTTP/1.1
14	4.680920	128.119.245.12	192.168.1.102	HTTP	490	HTTP/1.1 200 OK (text/html)

Packet number 14 contains the status code and phrase ('200 OK') associated with the response to the HTTP GET request.

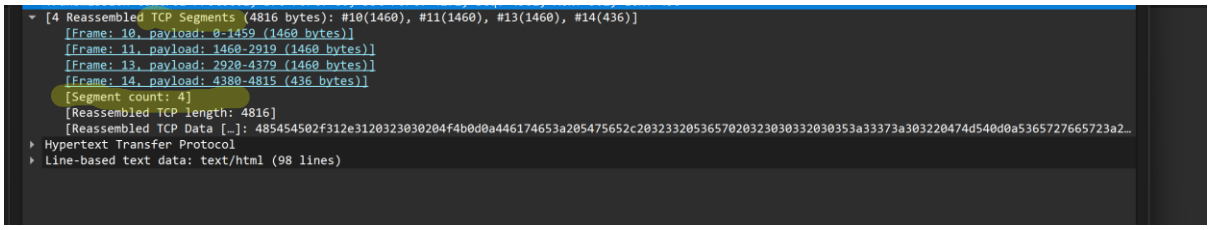
13. What is the status code and phrase in the response?



No.	Time	Source	Destination	Protocol	Length	Info
8	4.623732	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-3.html HTTP/1.1
14	4.680920	128.119.245.12	192.168.1.102	HTTP	490	HTTP/1.1 200 OK (text/html)

200 OK.

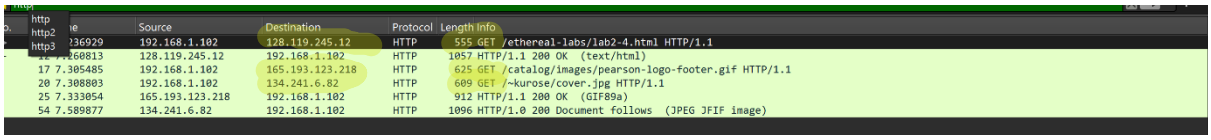
14. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?



```
[4 Reassembled TCP Segments (4816 bytes): #10(1460), #11(1460), #13(1460), #14(436)]
    [Frame: 10, payload: 0-1459 (1460 bytes)]
    [Frame: 11, payload: 1460-2919 (1460 bytes)]
    [Frame: 13, payload: 2920-4379 (1460 bytes)]
    [Frame: 14, payload: 4380-4815 (436 bytes)]
    [Segment count: 4]
    [Reassembled TCP length: 4816]
    [Reassembled TCP Data […]: 485454502f312e3120323030204f4b0d0a446174653a205475652c203233205365702023030332030353a33373a303220474d540d0a5365727665723a2…
Hypertext Transfer Protocol
Line-based text data: text/html (98 lines)
```

Segment count: 4

15. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?



No.	Time	Source	Destination	Protocol	Length	Info
[illegible]	[illegible]36929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
[illegible]	[illegible]60813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

3 GET Requests

128.119.245.12

165.193.123.218

134.241.6.82

16.Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

No.	Time	Source	Destination	Protocol	Length	Info
	236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
	260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

By examining the timestamps above in the packet list, it is evident that the GET requests for the two images were sent almost simultaneously (at 7.305485 and 7.308803) before the response for the first image was fully received. This overlap indicates that the browser downloaded the images in parallel rather than sequentially.

17. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

No.	Time	Source	Destination	Protocol	Length	Info
6	2.508229	192.168.1.102	128.119.245.12	HTTP	571	GET /ethereal-labs/protected_pages/lab2-5.html HTTP/1.1
9	2.538231	128.119.245.12	192.168.1.102	HTTP	278	HTTP/1.1 401 Authorization Required (text/html)
65	18.516793	192.168.1.102	128.119.245.12	HTTP	622	GET /ethereal-labs/protected_pages/lab2-5.html HTTP/1.1
68	18.541671	128.119.245.12	192.168.1.102	HTTP	499	HTTP/1.1 200 OK (text/html)

401 Authorization Required

18. When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message ?

```
GET /ethereal-labs/protected_pages/lab2-5.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /ethereal-labs/protected_pages/lab2-5.html
    Request Version: HTTP/1.1
  Host: gaia.cs.umass.edu\r\n
  User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
  Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,text/css,*/*;q=
  Accept-Language: en-us, en;q=0.50\r\n
  Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
  Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
  Keep-Alive: 300\r\n
  Connection: keep-alive\r\n
  Authorization: Basic ZXRoLXN0dWRlbnRzOm5ldHdvcmtz\r\n
  \r\n
  [Response in frame: 68]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/protected_pages/lab2-5.html]
```

A new 'Authorization' field is included in the second HTTP GET message. This is included because the requested page is password-protected. The browser encodes the username and password in Base64 format and sends them within this header to authenticate with the server and gain access to the file.